Omar Al-Zoubi

AI Engineer | Generative AI | LLMs | RAG & Agentic AI
Saudi Arabia

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Summary

AI Engineer with hands-on experience building and deploying Generative AI, LLM, RAG, agentic, multimodal, and computer vision systems. Experienced in model fine-tuning, prompt engineering, retrieval pipelines, LLM routing, and AI deployment using Python, PyTorch, Hugging Face, FastAPI, Docker, FAISS, Ollama, and vLLM. Focused on building production-oriented AI systems with optimized latency, inference cost, and reliability.

PROFESSIONAL EXPERIENCE

Co-Founder & Generative AI Engineer | Mabna Tech | Nov 2025 – Sep 2026

- Architected a configurable AI customer-service platform with domain-specific workflows for multiple business use cases.
- Designed a modular LLM architecture that dynamically loads only task-relevant prompts and context to reduce token usage, inference cost, and response latency.
- Built an end-to-end multilingual voice AI pipeline integrating LLMs, STT, TTS, automatic language detection, and Twilio telephony.
- Developed client-server AI services using FastAPI, Docker, and SQL to support scalable deployment and low-latency workflows.

AI Engineer Intern | KABi | Jun 2025 – Oct 2025

- Built an AI HR retrieval system with structured filtering and LLM reasoning, benchmarking ~132 ms across 100K synthetic profiles.
- Fine-tuned and deployed a multilingual XLM-R classifier to route user queries across specialized AI workflows.
- Designed adaptive query processing to bypass unnecessary LLM calls for simple requests while reserving deeper reasoning for complex queries.

Data Analyst | Al-Sham Province Hotels | Mar 2023 – May 2025

- Developed computer vision prototypes for hotel operations, including vehicle detection and parking monitoring using object-detection models.
- Built dashboards and KPI reports to track operational performance, financial trends, and business metrics.

SELECTED PROJECTS & RESEARCH

Team Lead – NOOR Smart Glasses | Graduation Project, UPM | 2024–2025

- Led development of AI-powered smart glasses for blind users with scene understanding, hazard detection, and contextual memory.
- Built a near-real-time client-server AI architecture with 0.3–1.1s scene-understanding response times.
- Trained a YOLOv8 hazard-detection model achieving 87% mAP@0.5 across critical hazard classes.
- Implemented fully local Arabic–English voice interaction and visual Q&A without cloud dependency.

Multi-Armed Bandit LLMs for Mathematical Reasoning | Research Paper | 2025–2026

- Co-developed an adaptive dual-LLM framework using an ε-greedy bandit to select among four prompting strategies.
- Improved relative accuracy by 22% while reducing average latency by 47% versus a fixed Chain-of-Thought baseline.

MizanAI – AI Personal Finance Assistant | 2026

- Designed a modular GenAI architecture combining intent routing, transaction extraction, temporal analysis, and personalized financial reasoning.
- Implemented dynamic prompt orchestration and context gating to reduce token overhead, unnecessary LLM calls, and inference cost.

Recommendation System | Assin Cafe | 2023–2024

- Built and deployed a collaborative-filtering recommendation system integrated with the cafe’s POS workflow.
- Achieved an 80% recommendation match rate in real-world customer interactions.

Education

Bachelor of Artificial Intelligence Engineering | University of Prince Muqrin (UPM) | 2020–2025

Technical Skills

- Programming: Python, Java, JavaScript
- AI & Machine Learning: LLMs, Generative AI, RAG, Agentic AI, Machine Learning, Deep Learning, Computer Vision, LLMs fine-tuning
- Frameworks & Models: PyTorch, Hugging Face Transformers, FAISS, Ollama, vLLM, XLM-RoBERTa, YOLO
- Backend & APIs: FastAPI, REST APIs, Client-Server Architecture
- Databases: SQL, PostgreSQL, SQLite, MongoDB
- Cloud & Deployment: AWS, Docker, Git, Linux

Certifications

- AWS Certified Solutions Architect – Associate | AWS | 2026
- Microsoft Generative AI Engineering Professional Certificate | Microsoft | 2026
- IBM Generative AI Engineering Professional Certificate | IBM | 2025
- IBM RAG & Agentic AI Professional Certificate | IBM | 2025
- Microsoft DevOps Engineering Professional Certificate | Microsoft | 2026
- Microsoft Power BI Data Analyst Professional Certificate | Microsoft | In Progress

Honors & Leadership

- Founder, Artificial Intelligence Students Club | UPM | 2022
- Contributor, ABET Accreditation | College of Computer Science | 2026